

# Matthew (Moslem) Shokrolahi

Toronto, ON | [moslem.sh.99@gmail.com](mailto:moslem.sh.99@gmail.com) | (613) 449-7401 | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

## SUMMARY

Machine Learning Engineer with a Ph.D. in Computer Engineering and 7+ years of experience designing, fine-tuning, and deploying AI applications powered by deep learning and large language models. Hands-on experience building LLM-enabled RAG systems and fine-tuning large pre-trained models, including LLMs and VLMs, supported by a strong software engineering foundation in Python, MLOps, and CI/CD.

## CORE EXPERTISE

LLM Applications & Fine-Tuning | RAG & Retrieval Systems | Computer Vision | Deep Learning | Generative AI | Diffusion Models | MLOps | AWS | Continual Learning | Responsible AI / Fairness & Bias

## PROFESSIONAL EXPERIENCE

**Applied Machine Learning Scientist: Model Risk Management & Compliance Oversight** | [TD Bank](#) | Toronto, ON | Jan 2026 - Present

- Research and apply machine-learning and deep-learning techniques to large-scale banking datasets, evaluating predictive performance, stability, explainability, and model behavior.
- Develop production-ready ML evaluation pipelines for fraud, insurance, and marketing models, translating research methodologies into scalable tooling in collaboration with engineering and model-development teams.
- Research evaluation methodologies for LLM, VLM, and agentic AI systems, including bias/fairness, explainability, robustness, and model-behavior assessment.

**AI Consultant** | [KnowQuest](#) | Toronto, ON | Sep 2023 - Sep 2025

- Developed a CLIP-based approach to detect AI-generated or mismatched review images using image-text alignment, improving fake review detection precision by 18%.
- Enhanced the feedback and recommendation system with an LLM-enabled RAG framework, improving mean reciprocal rank (MRR) by 23% and strengthening recommendation relevance.

**Research Assistant** | [UAI Lab, Queen's University](#) | Kingston, ON | May 2021 - Jul 2026

- Developed interpretable and controllable methods for safe and fair text-to-image diffusion, improving demographic fairness while maintaining image quality and background consistency.
- Developed continual learning methods using deep metric learning to reduce catastrophic forgetting and inter-task confusion; also investigated zero-shot machine unlearning, training models on GPU-accelerated HPC clusters (SLURM).

**Machine Vision Engineer** | [SAESPADAN](#) | Isfahan | Sep 2018 - May 2021

Designed a real-time stone-classification vision system using an ensemble of ResNet and EfficientNet models, reducing human error by 40% and increasing throughput by about 30% through weighted model averaging

## TECHNICAL SKILLS

**Programming:** Python (expert), SQL, C++ (proficient), C# (proficient), MATLAB

**Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, Pandas, OpenCV

**Software Systems & ML Engineering:** AWS, GPU-accelerated training, SLURM/HPC, Docker, FastAPI, REST APIs, MLflow, FAISS/vector search, Git/GitHub, GitHub Actions/CI/CD, Linux, Jupyter, LangChain, LangGraph

## EDUCATION

**Ph.D. in Computer Engineering** | [Queen's University, Kingston, ON](#) | May 2021 - Jul 2026

*Dissertation: Adaptive and Responsible Machine Learning: From Incremental Learning to Unlearning and Safe Generation*

**M.Sc. in Computer Engineering** | [Isfahan University of Technology \(IUT\), Isfahan](#) | Sep 2015

## INDEPENDENT PROJECT AND PROFESSIONAL DEVELOPMENT

**Smart Resume Screening:** Developing a resume-screening workflow that ingests PDF/DOCX resumes and technical job descriptions, extracts structured skills, experience, education, and certifications, and separates identity information from ranking features.

**Agentic AI Evaluation 1.0 Bootcamp** | [Vector Institute](#) | Feb - Apr 2026

## SELECTED PUBLICATIONS

- Shokrolahi, Sayedmoslem, Jae-Mo Kang, and Il-Min Kim. "Responsible Text-to-Image Diffusion: Interpretable and Linearly Controllable Semantics for Fair and Safe Generation." Forty-third International Conference on Machine Learning (ICML), 2026.
- Shokrolahi, Sayedmoslem, Jae-Mo Kang, and Il-Min Kim. "Learning Where It Matters: Responsible and Interpretable Text-to-Image Generation with Background Consistency." Transactions on Machine Learning Research (TMLR), 2026.
- Shokrolahi, Sayedmoslem, and Il-Min Kim. "MaxDiv: Zero-Shot Machine Unlearning via Distributionally Divergent Erasing Samples." IEEE Transactions on Artificial Intelligence (TAI), 2025.
- Shokrolahi, Sayedmoslem, and Il-Min Kim. "Combating Inter-Task Confusion and Catastrophic Forgetting by Metric Learning and Re-Using a Past Trained Model." Transactions on Machine Learning Research (TMLR), 2025.